

ASSIGNMENT (2)

Part 1: True / False

1. In a plan-driven process, planning is incremental and it is easy to change the process to reflect changing customer requirements.

False

2. The waterfall model is only appropriate when requirements are well-understood and are not expected to change much during the design process.

True

3. A major benefit of incremental development is that the process is highly visible, making it easy for managers to measure progress.

False

4. The four basic process activities—specification, development, validation, and evolution—are organized in the same sequence in all software process models.

False

5. Software evolution is an inherently flexible process of changing the system in response to changing customer needs.

True

Part 2: Multiple Choice (MCQ)

1. A "structured set of activities required to develop a software system" is the definition of:

a) A software process model

b) The software process

c) A plan-driven process

d) The Rational Unified Process

2. Which software process model is a plan-driven model with separate and distinct phases of specification and development?

a) The waterfall model

b) Incremental development

c) Reuse-oriented software engineering

d) The agile model

3. Which of the following is NOT listed as a benefit of incremental development?

a) The cost of accommodating changing customer requirements is reduced.

b) It is easier to get customer feedback on the development work.

c) System structure is improved with each new increment.

d) Rapid delivery and deployment of useful software is possible.

4. The design activity that involves identifying the overall structure of the system, its principal components, and their relationships is called:

a) Interface design

b) Component design

c) Database design

d) Architectural design

5. The testing stage where individual components are tested independently is known as:

a) System testing

- b) Acceptance testing
- c) Component testing**
- d) Validation testing

Part 3: Essay Questions

1. Compare Process Models: Contrast the waterfall model with the incremental development model. According to the lecture, how does each model organize the core process activities (specification, development, validation)? Which model is better for accommodating changing customer requirements, and why?.

Waterfall: Linear process (Specification → Development → Validation). Each phase is done once; hard to change later.

Incremental: System built in small parts; each increment goes through specification, development, and validation.

Better for changing requirements: *Incremental model*, because it allows continuous feedback and easy updates after each increment.

2. Describe the Reuse-Oriented Model: Explain the reuse-oriented software engineering model. What is its fundamental concept? List the main process stages involved in this model and provide one example of a software component (from the slides) that could be reused.

Concept: Build new systems by reusing existing software components instead of creating everything from scratch.

Main stages:

1-Component analysis

2-Requirements modification

3-System design with reuse

4-Development and integration

5- validation

Example: Reusing a login/authentication module in multiple apps.